# PRODUCT INVENTORY MANAGEMENT SYSTEM

**A MINI-PROJECT REPORT**

***Submitted by***


| | |
|---|---|
| **ROKITH S** | **241901090** |
| **SRI RAAM P** | **241901111** |


***in partial fulfillment of the award of the degree***

***of***

**BACHELOR OF ENGINEERING**

**IN**

**COMPUTER SCIENCE AND ENGINEERING**

**(CYBER SECURITY)**


**RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI**

**An Autonomous Institute**


**NOVEMBER 2025**

## BONAFIDE CERTIFICATE

Certified that this project **“PRODUCT INVENTORY MANAGEMENT SYSTEM”** is the bonafide work of **“ROKITH S, SRI RAAM P”** who carried out the project work under my supervision.

**SIGNATURE**

**Ms.R.Rupmala**

**ASSISTANT PROFESSOR**

Department of Computer Science and Engineering(Cyber Security)
Rajalakshmi Engineering College
Chennai

This mini project report is submitted for the viva voce examination to be held on __________

**NTERNAL EXAMINER** **EXTERNAL EXAMINER**

# DECLARTION

We hereby declare that the mini project report **PRODUCT INVENTORY**, submitted as part of the curriculum requirements for the Bachelor of Engineering (B.E) degree affiliated to Anna University, is a bonafide work carried out by us under the supervision of Ms. R. Rupmala, Assistant Professor, Department of Computer Science Engineering and Cyber Security, Rajalakshmi Engineering College, Chennai.

This submission represents our ideas in our own words, and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact, or source in our submission. We understand that any violation of the above will be grounds for disciplinary action by the institute and/or the University and may also evoke penal action from the sources which have not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other University.

Rajalakshmi Engineering College, **ROKITH S**

Chennai **SRI RAAM P**

November 2025

# ABSTRACT

In our region, small and medium-sized businesses form the backbone of the local economy. Despite the presence of large-scale retail and e-commerce platforms that manage inventory with advanced systems, many local vendors still rely on manual methods that are prone to errors and inefficiencies. To address this gap, our team developed a product inventory database system tailored to the needs of local businesses. The primary objective of this project is to streamline inventory management by tracking product availability, categorizing items, and monitoring stock levels in real time. This system empowers vendors to organize their inventory efficiently, reduce wastage, and respond swiftly to customer demands. By adopting this solution, local businesses can enhance their operational accuracy and compete more effectively with larger enterprises.

## ACKNOWLEDGEMENT

We like to convey our sincere appreciation to all who have supported and mentored us during the successful completion of this project work.

We express our profound gratitude to **Mr. Benedict J.N**., Associate Professor (SG) and Head of the Department of Computer Science Engineering and Cyber Security at Rajalakshmi Engineering College, for furnishing us with essential resources, support, and an enabling environment to execute this project.

We express our profound gratitude to **Ms. R. Rupmala,** Assistant Professor in the Department of Computer Science Engineering and Cyber Security at Rajalakshmi Engineering College, for her unwavering support, invaluable insights, and collaboration throughout our endeavor.

We would like to convey our gratitude to our faculty members and colleagues for their valuable feedback and encouragement.

We express our gratitude to our families and friends for their steadfast support, patience and encouragement, which were instrumental in the effective execution of this seminar

**ROKITH S**

**SRI RAAM P**

# Table of Contents

# LIST OF FIGURES

# CHAPTER 1

# INTRODUCTION

## 1.1 Project Overview

The Product Inventory System is a desktop-based application designed to simplify the process of managing product stock, supplier details, and category-wise organization within a store or company. It enables users to efficiently track product inflow and outflow, monitor stock levels in real time, and maintain accurate records of all inventory operations.

The system is built using **JavaFX** for the graphical user interface, providing an intuitive and responsive experience, and **MySQL** as the database backend for secure and structured data storage. Each product's details—such as name, category, supplier, quantity, and pricing—are stored in a relational database that ensures data integrity and easy retrieval.

The primary goal of this project is to automate manual inventory tracking, minimize errors, and provide quick insights into available stock, low-stock alerts, and overall inventory performance. This system can be customized for retail stores, warehouses, or small-scale businesses, offering scalability and efficiency through its modular architecture.

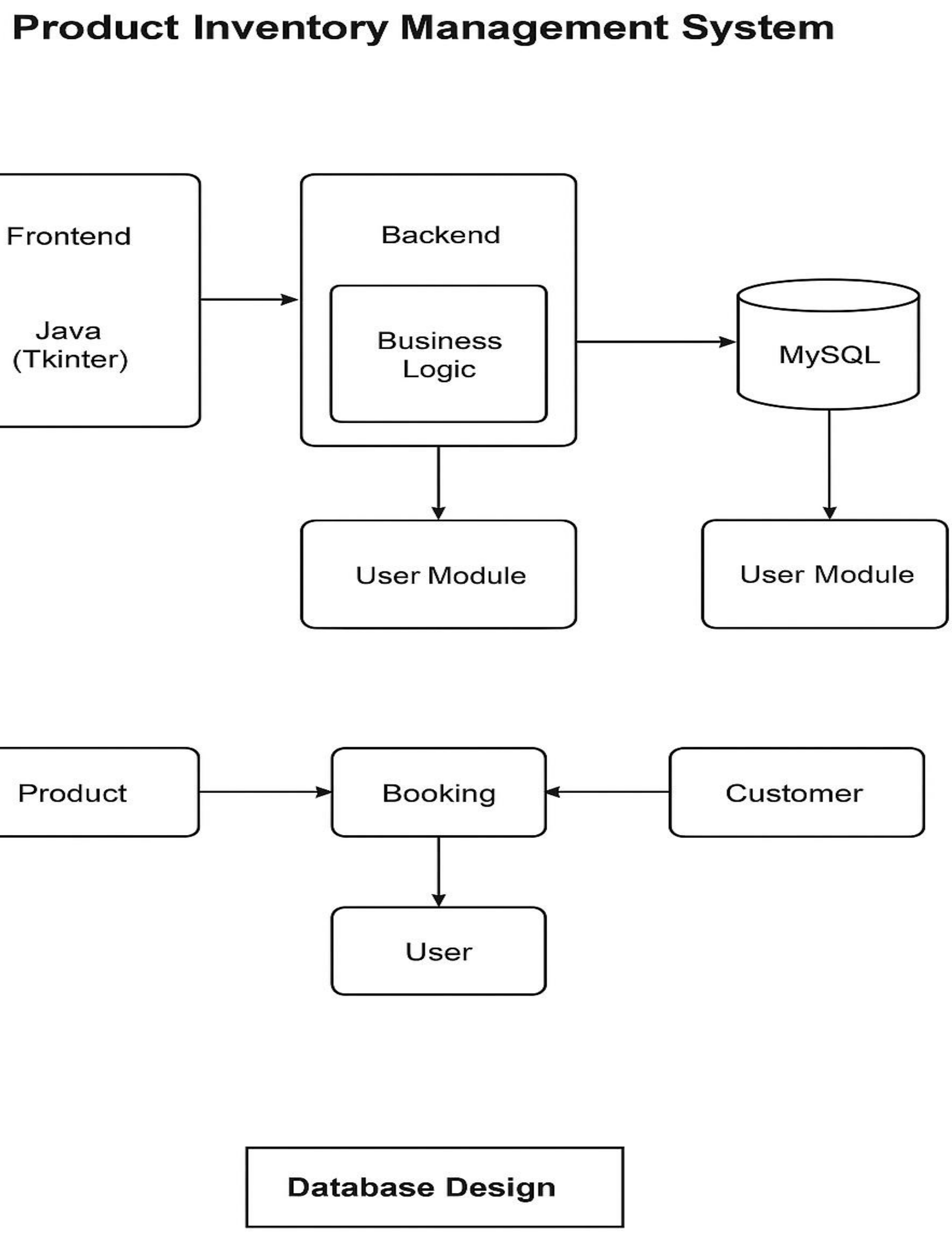



**FIG.1.1.DATABASE ARCHITECTURE**

## 1.2 Scope of the Work

The scope of this project is to develop a complete, standalone ProductInventory Management System for small and medium-scale businesses. It aims to simplify the management of products, suppliers, and stock levels while ensuring data accuracy and easy accessibility.

Key features include:

- Efficient storage and retrieval of product details such as name, category, supplier, quantity, and pricing information.
- Support for all core inventory operations including product addition, modification, deletion, and stock updates (CRUD operations).
- Real-time tracking of inventory levels to prevent overstocking and stock shortages.
- Generation of detailed inventory and sales reports for business analysis and decision-making.
- Provision of a modern, responsive, and user-friendly graphical interface (JavaFX) for smooth interaction.
- Integration with a secure and persistent MySQL database for reliable data management.
- Scalable architecture allowing future integration of advanced features such as barcode scanning, supplier management automation, and predictive restocking alerts.

## 1.3 Problem Statement

In today's competitive marketplace, efficient inventory management plays a vital role in the success of any business. However, many small and medium-scale enterprises lack access to advanced inventory management systems used by large corporations. As a result, they often rely on manual recordkeeping or outdated software, which leads to frequent issues such as stock mismanagement, overstocking, stockouts, and inaccurate sales tracking.

Without a proper digital system in place, vendors struggle to monitor product availability, reorder levels, and supplier information effectively. This lack of real-time visibility not only results in operational inefficiencies but also causes financial losses, delays in order fulfillment, and decreased customer satisfaction.

The Product Inventory Management System aims to bridge this technological gap by providing an affordable, user-friendly, and scalable software solution. It allows businesses to automate inventory operations, maintain accurate product data, and generate real-time reports for better decision-making. By ensuring proper stock control and data accessibility, this system empowers local vendors and mid-sized enterprises to compete effectively in a digitally-driven business environment.

## 1.4 Aim and Objectives of the Project

The main aim of this project is to develop a simple, efficient, and reliable Product Inventory Management System that helps businesses organize, monitor, and control their stock effectively. It ensures accurate tracking of products, categories, and quantities to maintain smooth business operations and prevent stock-related issues.

The specific objectives are:

- To maintain detailed and up-to-date records of all products and their quantities.
- To track stock levels and alert users when items are low or out of stock.
- To support easy addition, update, and deletion of product details.
- To categorize products for quick search and better organization.
- To generate clear and concise inventory reports for business analysis.

# CHAPTER 2

## SYSTEM SPECIFICATIONS

### 2.1 HARDWARE SPECIFICATIONS

| | | |
|---|---|---|
| Processor | : | Intel i5 |
| Memory Size | : | 8GB (Minimum) |
| HDD | : | 1 TB (Minimum) |

### 2.2 SOFTWARE SPECIFICATIONS

| | | |
|---|---|---|
| Operating System | : | WINDOWS 10 |
| Front – End | : | Java |
| Back - End | : | MySql |
| Language | : | Java,SQL |

# CHAPTER 3

## MODULE DESCRIPTION

### 1. Product Management Module

**Purpose:**
Handles all product-related information including product ID, name, category, unit price, and quantity available.

**Key Features:**

- Add, update, and delete product details
- Maintain product categories and descriptions
- Assign unique product identifiers
- Ensure data integrity using primary and foreign keys

### 2. Inventory Control Module

**Purpose:**
Tracks stock levels, manages reorder points, and updates quantities after sales or restocks.

**Key Features:**

- Monitors available quantity in real time
- Automatically flags low-stock products
- Updates inventory after purchase and sales transactions
- Generates inventory status reports

### 3. Supplier Management Module

**Purpose:**
Stores and manages supplier information for restocking and procurement operations.

**Key Features:**

- Maintain supplier details (ID, name, contact, address)
- Link suppliers to the products they provide
- Enable supplier performance tracking and purchase history

**4. Purchase and Sales Module**

**Purpose:**
Records incoming stock (purchases) and outgoing stock (sales).

**Key Features:**

- Manage purchase orders and invoices
- Record customer and transaction details
- Automatically update stock after each transaction
- Generate purchase/sales summaries and reports

**5. Search and Reporting Module**

**Purpose:**
Facilitates quick data retrieval and provides analytical insights.

**Key Features:**

- Search products by ID, name, or category
- Generate reports (stock status, product value, sales summary)
- Export data for analysis or printing

**6. User Interface Module (optional, if using GUI)**

**Purpose:**
Provides an easy-to-use interface for database interaction, using forms and tables.

**Key Features:**

- Input validation and error handling

- Interactive forms for product, supplier, and transaction management
- Table views for search and reporting

**7. Database and Security Module**

**Purpose:**
Ensures secure, reliable storage and retrieval of data.

**Key Features:**

- Use of relational tables with constraints and relationships
- Backup and recovery mechanisms
- Controlled user access and authentication

# CHAPTER 4

## SAMPLE CODING

import javax.swing.*; import javax.swing.table.DefaultTableModel; import java.awt.*; import java.awt.event.*; import java.sql.*; import java.util.Vector;

public class SmartInventoryH2 extends JFrame { Connection conn; JTextField txtId, txtName, txtQty, txtPrice, txtSearch; JTable table; DefaultTableModel model;

```
public SmartInventoryH2() {
    setTitle("SmartInventory H2");
    setSize(800, 500);
    setDefaultCloseOperation(EXIT_ON_CLOSE);
    setLocationRelativeTo(null);
    initDB(); initUI(); loadData();
}

void initDB() {
    try {
        conn =
DriverManager.getConnection("jdbc:h2:./inventory_db",
"sa", "");
        Statement s = conn.createStatement();
        s.execute("""
```

```
            CREATE TABLE IF NOT EXISTS products(
                id INT AUTO_INCREMENT PRIMARY KEY,
                name VARCHAR(100), quantity INT
CHECK(quantity>=0),
                price DECIMAL(10,2) CHECK(price>0),
                last_updated TIMESTAMP DEFAULT
CURRENT_TIMESTAMP)
        """);
        ResultSet rs = s.executeQuery("SELECT COUNT(*)
FROM products"); rs.next();
        if (rs.getInt(1)==0)
            s.execute("INSERT INTO
products(name,quantity,price)VALUES" +

"('Laptop',8,850),('Mouse',45,25.5),('Keyboard',3,120
)");
    } catch (SQLException e) { showError(e);
System.exit(0); }
}

void initUI() {
    JPanel top = new JPanel(new BorderLayout());
    JPanel search = new JPanel();
    search.add(new JLabel("Search:"));
    txtSearch = new JTextField(20); JButton btnSearch
= new JButton("Search");
    search.add(txtSearch); search.add(btnSearch);
```

```
top.add(search, BorderLayout.NORTH);

    JPanel input = new JPanel(new GridLayout(4, 2, 10,
10));

input.setBorder(BorderFactory.createTitledBorder("Pro
duct Details"));
    input.add(new JLabel("ID:")); input.add(txtId =
new JTextField());
    input.add(new JLabel("Name:")); input.add(txtName
= new JTextField());
    input.add(new JLabel("Qty:")); input.add(txtQty =
new JTextField());
    input.add(new JLabel("Price:"));
input.add(txtPrice = new JTextField());
    top.add(input, BorderLayout.CENTER);

    JPanel btns = new JPanel();
    String[] ops =
{"Add","Update","Delete","Clear","Refresh"};
    JButton[] b = new JButton[ops.length];
    for (int i=0;i<ops.length;i++){ b[i]=new
JButton(ops[i]); btns.add(b[i]); }
    top.add(btns, BorderLayout.SOUTH); add(top,
BorderLayout.NORTH);

    table = new JTable(); add(new JScrollPane(table));
```

```
table.setRowHeight(25);
    b[0].addActionListener(e->add());
b[1].addActionListener(e->update());
    b[2].addActionListener(e->delete());
b[3].addActionListener(e->clear());
    b[4].addActionListener(e->loadData());
btnSearch.addActionListener(e->search());
    txtSearch.addKeyListener(new KeyAdapter(){ public
void    keyReleased(KeyEvent    e){    search();    }});
    table.addMouseListener(new        MouseAdapter(){
        public  void  mouseClicked(MouseEvent  e){  int
r=table.getSelectedRow();
            if(r>=0){
txtId.setText(model.getValueAt(r,0).toString());

txtName.setText(model.getValueAt(r,1).toString());

txtQty.setText(model.getValueAt(r,2).toString());

txtPrice.setText(model.getValueAt(r,3).toString()); }
}});
}

void                   loadData()                    {
    try    (Statement    s=conn.createStatement())    {
        updateTable(s.executeQuery("SELECT    *    FROM
products"));
```

```
    } catch (Exception e) { showError(e); }
}

void add() {
    if (!valid()) return;
    exec("INSERT INTO
products(name,quantity,price)VALUES(?,?,?)", false);
}

void update() {
    if (txtId.getText().isEmpty() || !valid()) return;
    exec("UPDATE products SET
name=?,quantity=?,price=? WHERE id=?", true);
}

void delete() {
    if (txtId.getText().isEmpty()) return;
    if (JOptionPane.showConfirmDialog(this,"Delete ID
"+txtId.getText()+"?","Confirm",

JOptionPane.YES_NO_OPTION)!=JOptionPane.YES_OPTION)
return;
    try (PreparedStatement
ps=conn.prepareStatement("DELETE FROM products WHERE
id=?")){

ps.setInt(1,Integer.parseInt(txtId.getText()));
```

```
JOptionPane.showMessageDialog(this,ps.executeUpdate()
>0?"Deleted!":"Not                             found!");
        clear();                               loadData();
    }    catch(Exception    e){    showError(e);    }
}

void                    search()                    {
    try                             (PreparedStatement
ps=conn.prepareStatement("SELECT * FROM products WHERE
name                    LIKE                    ?")){
        ps.setString(1,"%"+txtSearch.getText()+"%");
updateTable(ps.executeQuery());
    }    catch(Exception    e){    showError(e);    }
}

void    exec(String    sql,    boolean    upd)    {
    try                             (PreparedStatement
ps=conn.prepareStatement(sql)){
        ps.setString(1,txtName.getText());

ps.setInt(2,Integer.parseInt(txtQty.getText()));

ps.setDouble(3,Double.parseDouble(txtPrice.getText())
);
        if(upd)
ps.setInt(4,Integer.parseInt(txtId.getText()));
```

```
        ps.executeUpdate();

JOptionPane.showMessageDialog(this,upd?"Updated!":"Ad
ded!");
        int     q=Integer.parseInt(txtQty.getText());
        if(q<5)
JOptionPane.showMessageDialog(this,"LOW        STOCK:
"+q,"Alert",2);
        clear();                             loadData();
    }    catch(Exception    e){    showError(e);    }
}

boolean                   valid()                   {
    try                                             {
        int     q=Integer.parseInt(txtQty.getText());
        double
p=Double.parseDouble(txtPrice.getText());
        if(txtName.getText().isEmpty()||q<0||p<=0)
throw                 new                Exception();
        return                                  true;
    }               catch(Exception               e){
JOptionPane.showMessageDialog(this,"Invalid input!");
return                    false;                    }
}

void  updateTable(ResultSet  rs)throws  SQLException{
    model=new                    DefaultTableModel();
```

```
    ResultSetMetaData             m=rs.getMetaData();
    for(int            i=1;i<=m.getColumnCount();i++)
model.addColumn(m.getColumnName(i));
    while(rs.next()){      Vector<Object>      v=new
Vector<>();
        for(int        i=1;i<=m.getColumnCount();i++)
v.add(rs.getObject(i));
        model.addRow(v);                             }
    table.setModel(model);
}

void clear(){ txtId.setText(""); txtName.setText("");
txtQty.setText("");
    txtPrice.setText("");  txtSearch.setText("");   }

void             showError(Exception             e){
JOptionPane.showMessageDialog(this,e.getMessage(),"Er
ror",0);                                         }

public    static    void    main(String[]    a){
SwingUtilities.invokeLater(()->new
SmartInventoryH2().setVisible(true));            }


}
```

# CHAPTER 5

## SCREEN SHOTS

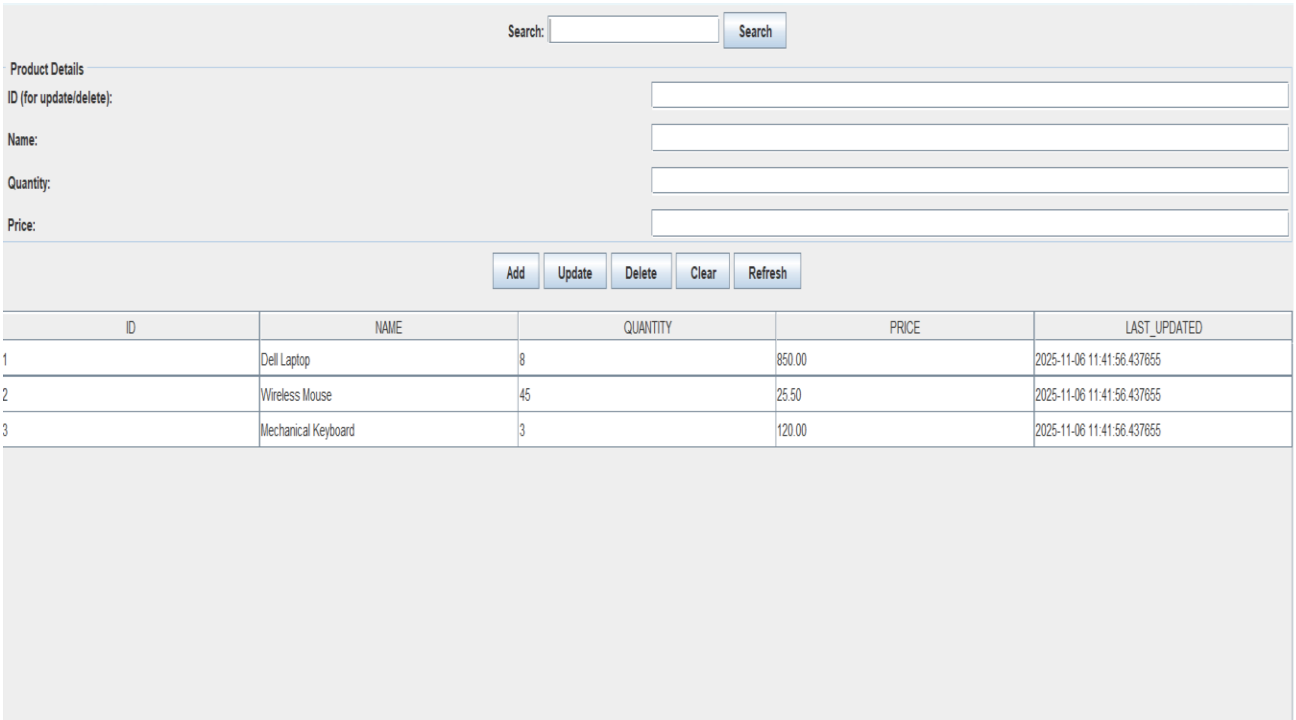



**Fig 5.1 USER INTERFACE**

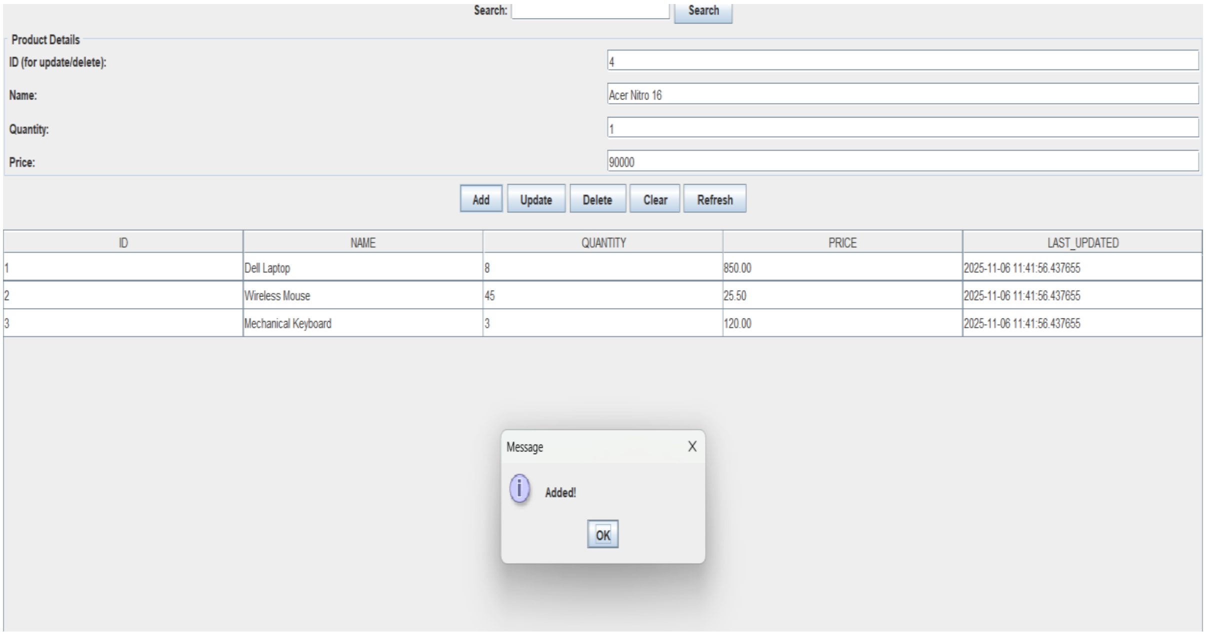



**Fig 5.2 ADDING A PRODUCT**

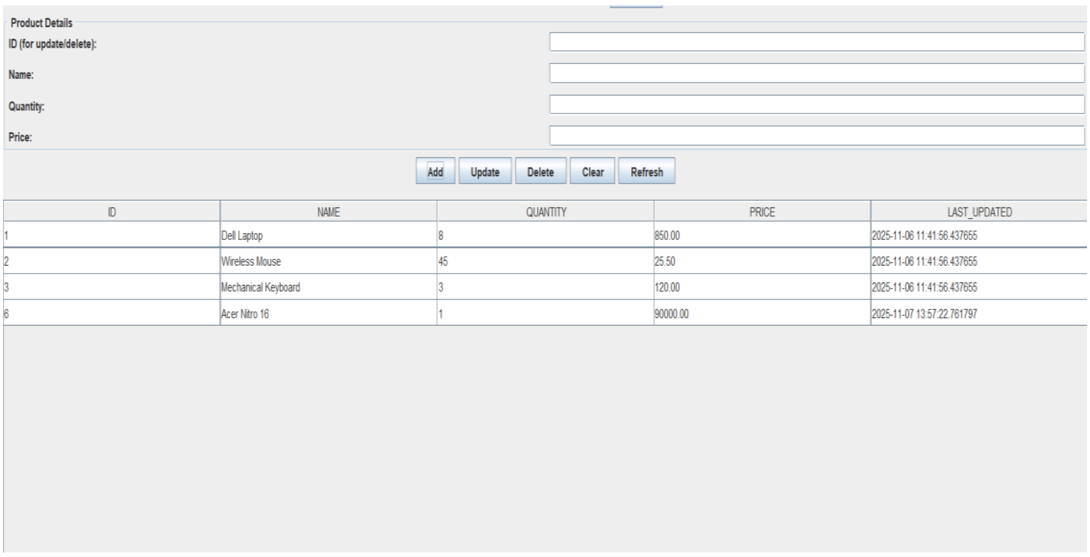
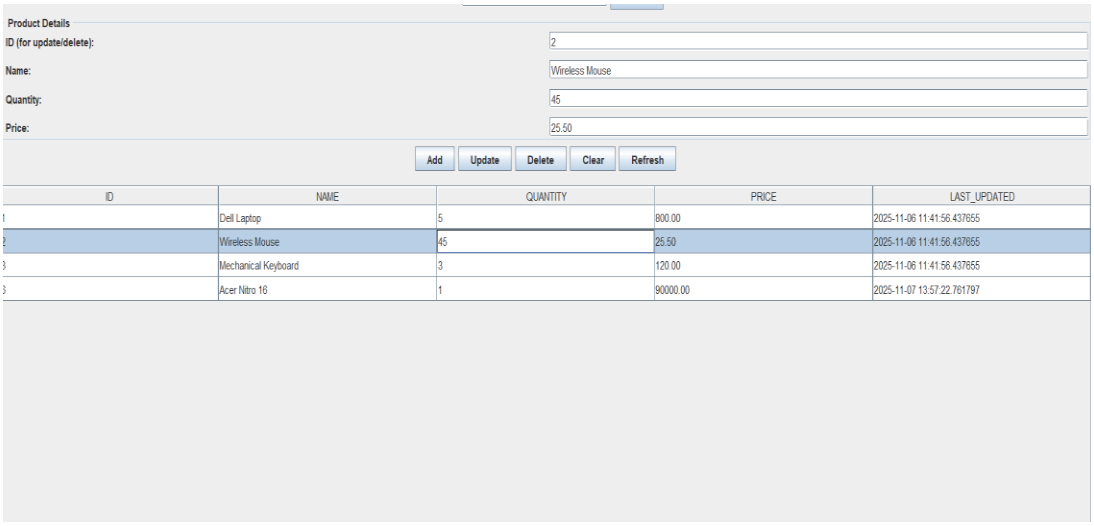
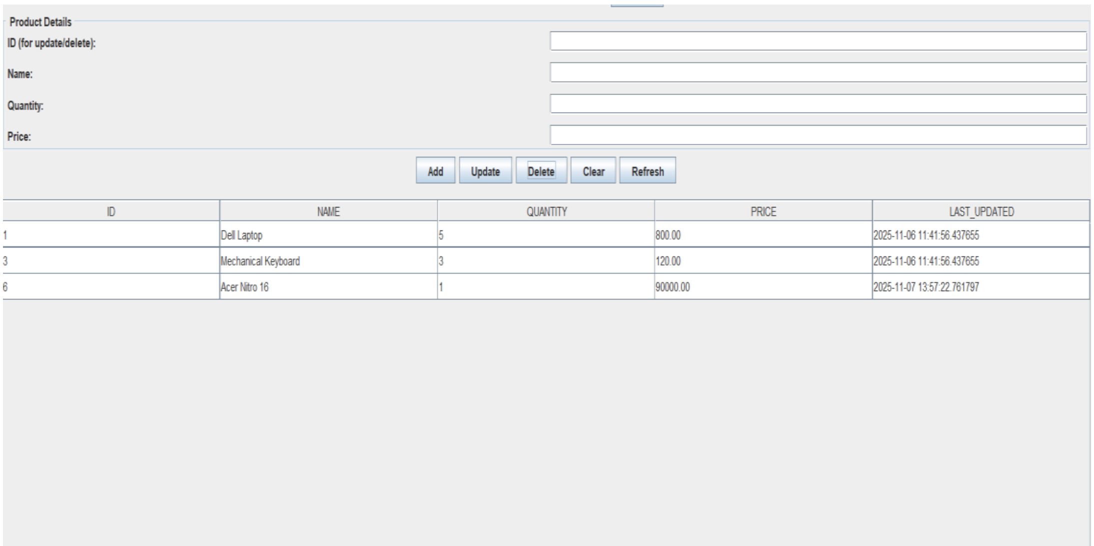



**Fig 5.3  AFTER ADDING A PRODUCT**




## Fig 5.4 DELETING A PRODUCT




## Fig 5.5 AFTER DELETING A PRODUCT

# CHAPTER 6

# CONCLUSION AND FUTURE ENHANCEMENT

## 6.1 Conclusion

The **Product Inventory Management System** successfully provides a reliable, efficient, and user-friendly solution for managing business inventory operations. It allows vendors and business owners to maintain detailed product information, track stock levels, and update inventory in real time. Through structured inventory logs and organized product categorization, the system reduces manual effort, minimizes stock-related errors, and improves overall operational accuracy.

By digitizing the inventory management process, the project helps businesses gain greater visibility into their stock flow and restocking requirements. This leads to faster decision-making, better customer satisfaction, and optimized use of resources. The system's modular design also ensures flexibility, enabling it to be adapted to various business scales and operational needs.

Overall, the Product Inventory Management System enhances productivity, accuracy, and accessibility — empowering local and mid-sized enterprises to compete effectively in today's technology-driven market.

## 6.2 Future Enhancements

The system is designed to be scalable and can be extended with advanced features in the future to enhance usability and automation.

**Possible future enhancements include:**

- **Automated Stock Updates:** Integration of sensors or POS systems to automatically update stock levels after each sale or purchase.

- **Low-Stock Notifications:** Real-time alerts and reminders when product quantities fall below the threshold level.

- **Supplier Management Module:** Automated supplier tracking and purchase order generation for better supply coordination.

- **Sales Platform Integration:** Connecting the system with e-commerce or billing platforms for live inventory synchronization.

- **Data Analytics Dashboard:** Visual reports and charts to analyze product trends, sales performance, and demand forecasting.

- **Multi-User Access Control:** Allowing multiple authorized users with role-based permissions for improved team collaboration.

# REFERENCES

1. https://www.w3schools.com/sql/
2. https://www.tutorialspoint.com/sqlite/index.htm
3. https://www.wikipedia.org/
4. https://www.learnpython.org/
5. https://www.codecademy.com/learn/learn-python